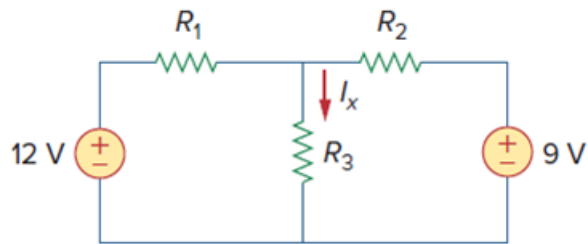


Problem

Using Fig. 3.50 from Prob. 3.1, design a problem to help other students better understand mesh analysis.

Figure 3.50:



Step-by-step solution

Step 1 of 6

Draw the circuit diagram shown in Figure 1.

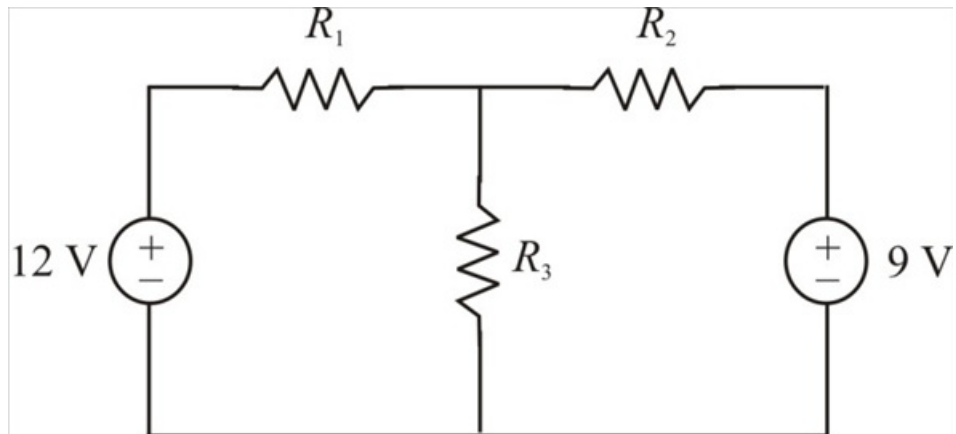


Figure 1

Step 2 of 6

The circuit is labeled as shown in Figure 2.

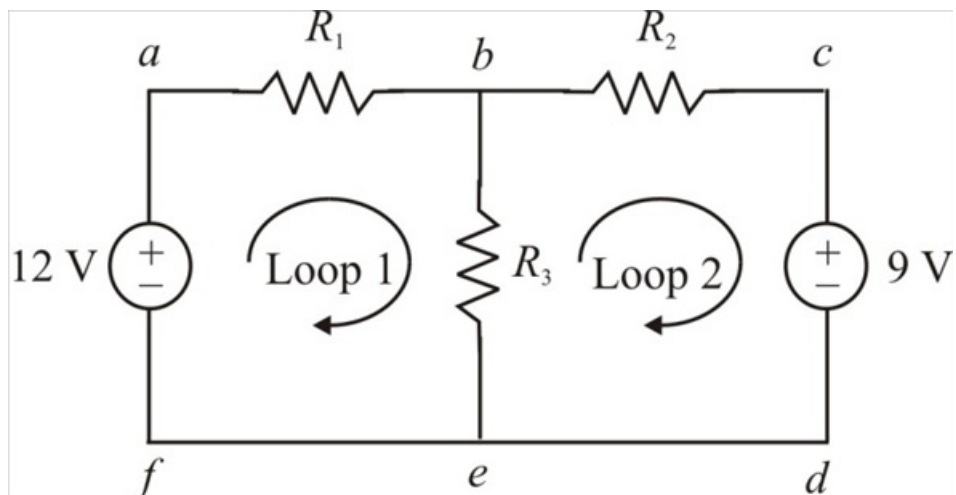


Figure 2

Step 3 of 6

In Figure 2, the path abefa and bcdeb are two meshes, but the path abcdefa is not a mesh. The current through the mesh is known as mesh current.

Let the mesh currents are i_1 and i_2 for loops 1 and 2 respectively.

In mesh analysis, we are interested in applying Kirchhoff's voltage law to find the mesh currents in the circuit.

Kirchhoff's voltage law states that the algebraic sum of all voltages around a closed path (or loop) is zero.

Applying Kirchhoff's voltage law for the loop 1, we get

$$-12 + i_1 R_1 + R_3(i_1 - i_2) = 0$$

$$(R_1 + R_3)i_1 - R_3i_2 = 12$$

$$\text{Therefore } (R_1 + R_3)i_1 - R_3i_2 = 12 \dots\dots (1)$$

Applying Kirchhoff's current law for the loop 2, we get

$$R_3(i_2 - i_1) + i_2 R_2 + 9 = 0$$

$$-R_3i_1 + (R_2 + R_3)i_2 = -9$$

$$\text{Therefore } -R_3i_1 + (R_2 + R_3)i_2 = -9 \dots\dots (2)$$

Step 4 of 6

If we know the values of three resistors, then equation (1) and (2) have two variables. We can solve the two equations for two variables i_1 and i_2 .

Let us assuming that,

$$R_1 = 2 \Omega$$

$$R_2 = 4 \Omega$$

$$R_3 = 6 \Omega$$

Substituting resistor values in equation (1), we get

$$(R_1 + R_3)i_1 - R_3i_2 = 12$$

$$(2 + 6)i_1 - 6i_2 = 12$$

$$8i_1 - 6i_2 = 12$$

$$4i_1 - 3i_2 = 6$$

$$\text{Therefore } 4i_1 - 3i_2 = 6 \dots\dots (3)$$

Step 5 of 6

Substituting resistor values in equation (2), we get

$$-R_3i_1 + (R_2 + R_3)i_2 = -9$$

$$-6i_1 + (4 + 6)i_2 = -9$$

$$-6i_1 + 10i_2 = -9$$

$$\text{Therefore } -6i_1 + 10i_2 = -9 \dots\dots (4)$$

Multiplying equation (3) by 6 and equation (4) by 4 and adding both, we get

$$24i_1 - 18i_2 = 36$$

$$\underline{-24i_1 + 40i_2 = -36}$$

$$22i_2 = 0$$

$$i_2 = 0 \text{ A}$$

Step 6 of 6

Substituting $i_2 = 0$ in equation (3), we get

$$4i_1 - 3i_2 = 6$$

$$4i_1 - 3(0) = 6$$

$$i_1 = 1.5 \text{ A}$$

The current I_x can be calculated as:

$$I_x = i_1 - i_2$$

$$= 1.5 - 0$$

$$= 1.5 \text{ A}$$

Therefore the current I_x is 1.5 A.